電子學實驗 (一)
Integrators, Differentiators, and Their Frequency Responses
預報

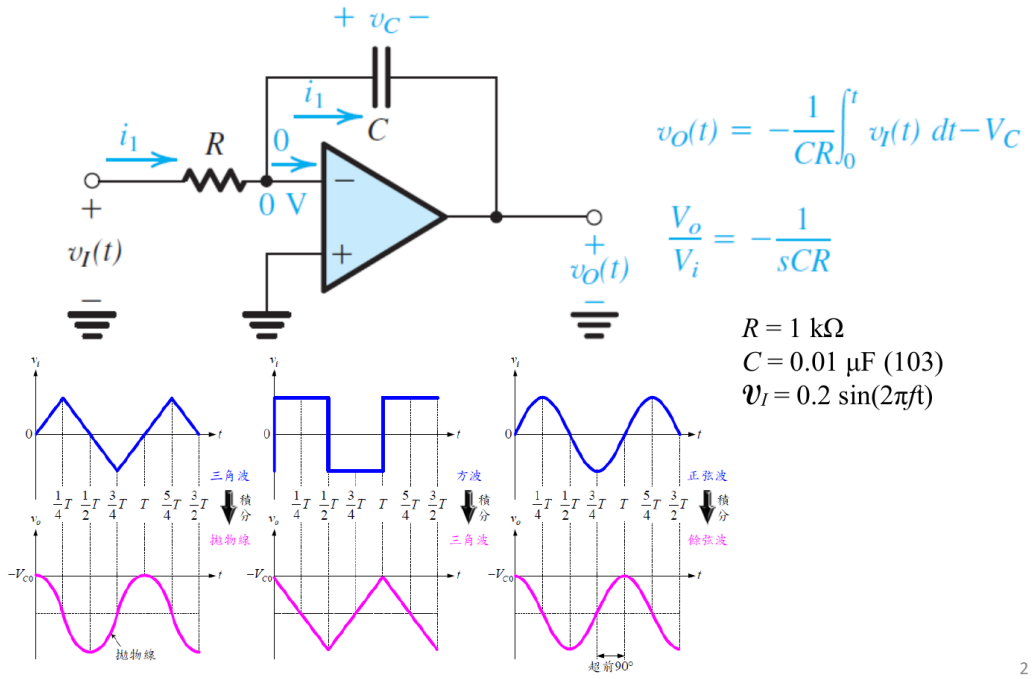
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1 實驗目的

- To understand the operating principles of Miller integrator and its feasible configuration with feedback resistor in parallel connection, and differentiator circuits.
- To observe their output waveform under triangle, square, and sinusoidal waves.
- To measure their frequency responses and sketch the Bode magnitude plots.

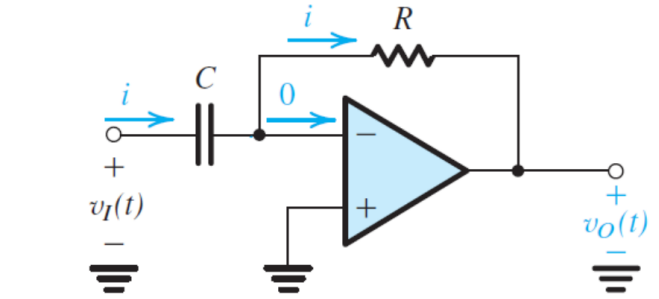
2 預備實驗的電路圖

Exp. 1 Integrator



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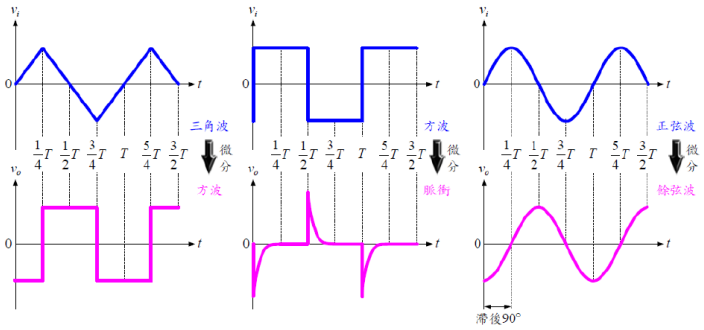
Exp. 2 Differentiator



$$i(t) = C\frac{dv_I(t)}{dt}$$

$$v_O(t) = -CR\frac{dv_I(t)}{dt}$$

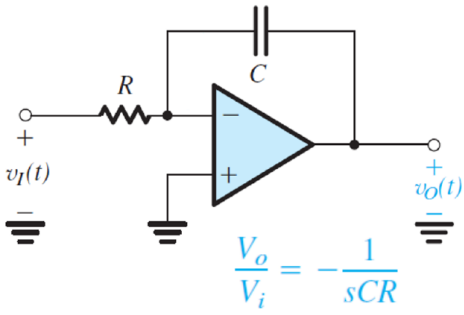
$$\frac{V_o}{V_i} = -sCR$$



$R = 1$ kΩ
$C = 0.01$ μF (103)
$v_I = 0.2\sin(2\pi ft)$

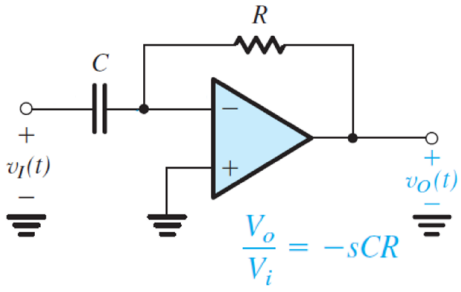
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Exp. 3 Frequency Response (I)



$$\frac{V_o}{V_i} = -\frac{1}{sCR}$$

$R = 1$ kΩ
$C = 0.01$ μF (103)
$v_I = 0.2\sin(2\pi ft)$, $f = 100$ Hz ~ 300 kHz
Please sketch the Bode magnitude plot.

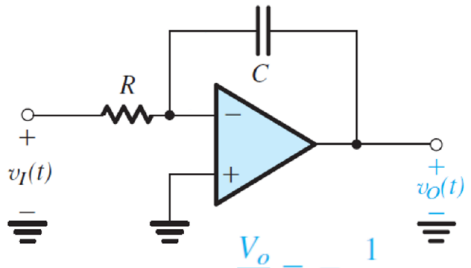


$$\frac{V_o}{V_i} = -sCR$$

$R = 1$ kΩ
$C = 0.01$ μF (103)
$v_I = 0.2\sin(2\pi ft)$, $f = 100$ Hz ~ 100 kHz
Please sketch the Bode magnitude plot.

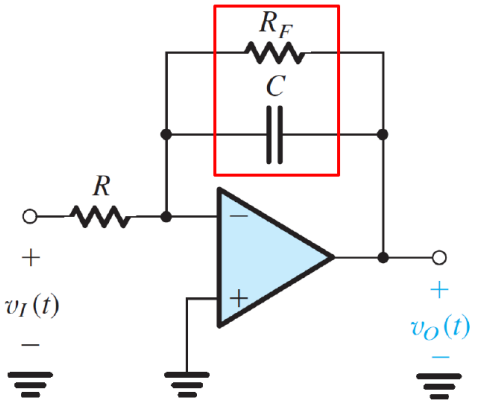
4

Exp. 3 Frequency Response (II)



$\frac{V_o}{V_i} = -\frac{1}{sCR}$

$R = 1$ kΩ
$C = 0.001$ μF (102)
$v_I = 0.2 \sin(2\pi ft), f = 100$ Hz ~ 1 MHz
Please sketch the Bode magnitude plot.



$R = 1$ kΩ, $R_F = ?$ kΩ
$C = 0.001$ μF (102)
$v_I = 0.2 \sin(2\pi ft), f = 100$ Hz ~ 1 MHz
Please sketch the Bode magnitude plot.

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3 預備量測的數據

I Exp. 1 Integrator

Measure v_I and v_O waveforms of an op amp where input voltage v_I is connected to v_- through an input resistance $R = 1$ kΩ(current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a capacitance $C = 0.01$ μF (103) (current from v_- to v_O), under triangle, square, and sinusoidal v_I with amplitude 0.2.

Theoretical results:

$v_O(0^-)$ is the initial voltage across C (v_O relative to v_-).

Time domain:

$$v_+ = v_- = 0$$

$$i_I = \frac{v_I}{R} = i_C = -C\frac{\mathrm{d}v_O}{\mathrm{d}t}$$

$$v_O = v_O(0^-) - \frac{1}{RC}\int_0^t v_I(\tau)\,\mathrm{d}\tau$$

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_C = -CsV_O + Cv_O(0^-)$$

$$V_O = \frac{1}{s}v_O(0^-) - \frac{1}{RCs}V_I$$

Transfer function: Infinite DC gain. 90 degrees phase shift.

$$H(s) = \frac{V_o}{V_i} = -\frac{1}{RCs}$$

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 15.9 \text{ kHz}$$

Zero-state responses:

- Triangle wave input:

$$v_I(t) = 4A\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right), \quad A, T > 0$$

$$v_O(t) = -\frac{AT}{RC}\left(2s\,|s| - s\right), \quad s = \frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor$$

- Square wave input:

$$v_I(t) = 2A\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad A, T > 0$$

  Triangle wave output with amplitude $\frac{AT}{4RC}$

$$v_O(t) = -\frac{AT}{RC}\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right)$$

- Sinusoidal input:

$$v_I(t) = A\sin\left(\frac{t}{T}\right), \quad A, T > 0$$

  Sinusoidal output 90 degrees leading input with amplitude $\frac{AT}{RC}$.

$$v_O(t) = \frac{AT}{RC}\cos\left(\frac{t}{T}\right)$$

II Exp. 2 Differentiator

Measure v_I and v_O waveforms of an op amp where input voltage v_I is connected to v_- through an input capacitor $C = 0.01$ μF (103) (current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a resistor $R = 1$ kΩ(current from v_- to v_O), under triangle, square, and sinusoidal v_I with amplitude 0.2.

Theoretical results:

$v_I(0^-)$ is the initial voltage across C (v_I relative to v_-).

Time domain:

$$i_I = C\frac{\mathrm{d}v_I}{\mathrm{d}t} = i_f = \frac{-v_O}{R}$$

$$v_O = -RC\frac{\mathrm{d}v_I}{\mathrm{d}t}$$

Laplace domain:

$$I_I = CsV_I - Cv_I(0^-) = I_f = \frac{-V_O}{R}$$

$$V_O = -RCsV_I + RCv_I(0^-)$$

Transfer function: Infinite gain at $\omega = \infty$. -90 degrees phase shift.

$$H(s) = \frac{V_o}{V_i} = -RCs$$

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 15.9 \text{ kHz}$$

Responses:

- Triangle wave input:

$$v_I(t) = 4A\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right), \quad A, T > 0$$

  Square wave with amplitude $\frac{4ARC}{T}$

$$v_O(t) = -\frac{8ARC}{T}\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad \frac{2t}{T} \notin \mathbb{Z}$$

- Square wave input:

$$v_I(t) = 2A\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad A, T > 0$$

  Dirac delta sum output

$$v_O(t) = 2ARC\sum_{n\in\mathbb{Z}} \delta(t - nT) - \delta(t - \left(n + \frac{1}{2}\right)T)$$

- Sinusoidal input:

$$v_I(t) = A\sin\left(\frac{t}{T}\right), \quad A, T > 0$$

  Sinusoidal output 90 degrees lagging input with amplitude $\frac{ARC}{T}$.

$$v_O(t) = -\frac{ARC}{T}\cos\left(\frac{t}{T}\right)$$

III Exp. 3 Frequency Responses (I)

Measure frequency response of the circuit in Exp. 1 Integrator under

$$v_I = 0.2\sin(2\pi ft), \quad f = 100 \text{ Hz} \sim 300 \text{ kHz}$$

and sketch the Bode magnitude plot.

Measure frequency response of the circuit in Exp. 2 Differentiator under

$$v_I = 0.2\sin(2\pi ft), \quad f = 100 \text{ Hz} \sim 100 \text{ kHz}$$

and sketch the Bode magnitude plot.

IV Exp. 3 Frequency Responses (II)

Measure frequency response of the circuit in Exp. 1 Integrator with the capacitor C replaced with a capacitor $C = 0.001$ μF (102) under

$$v_I = 0.2\sin(2\pi f t), \quad f = 100 \text{ Hz} \sim 1 \text{ MHz}$$

and sketch the Bode magnitude plot.

Theoretical results:

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 159 \text{ kHz}$$

Measure frequency response of the circuit in Exp. 1 Integrator with the capacitor C replaced with a capacitor $C = 0.001$ μF (102) and a resistor R_F in parallel connection under

$$v_I = 0.2\sin(2\pi f t), \quad f = 100 \text{ Hz} \sim 1 \text{ MHz}$$

and sketch the Bode magnitude plot.

Theoretical results:

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_F = -\left(\frac{1}{R_F} + Cs\right)V_O + Cv_O(0^-) = -\frac{1 + R_F Cs}{R_F}V_O + Cv_O(0^-)$$

$$V_O = \frac{R_F C}{1 + R_F Cs}v_O(0^-) - \frac{\frac{R_F}{R}}{1 + R_F Cs}V_I$$

Transfer function: STC low-pass with cutoff frequency $\frac{1}{2\pi R_F C}$ and DC gain $\frac{R_F}{R}$.

$$H(s) = \frac{V_o}{V_i} = -\frac{\frac{R_F}{R}}{1 + R_F Cs}$$